

Supplemental Materials

The supplemental materials provide some details of the 2+1 flavor low energy effective field theory within the functional renormalization group approach, [Section S.1](#), preliminary results of temperature fluctuations calculated in QCD within fRG, [Section S.2](#), hyper-order fluctuations of temperature, [Section S.3](#), ratios of temperature fluctuation cumulants, [Section S.4](#), relation between the temperature T and the mean transverse momentum $\langle p_T \rangle$, [Section S.5](#).

S.1. 2+1 flavor low energy effective field theory within the fRG

In this appendix, we recapitulate the setup of the 2+1 flavor low energy effective field theory (LEFT) within the functional renormalization group used in this work. More details about the 2+1 LEFT can be found in [\[1\]](#). The effective action reads

$$\Gamma_k[\Phi] = \int_x \left\{ \bar{q} [\gamma_\mu \partial_\mu - \gamma_0 (\mu + igA_0)] q + h_k \bar{q} \Sigma_5 q + \text{tr}(\partial_\mu \Sigma \cdot \partial_\mu \Sigma^\dagger) + V_{\text{matt},k}(\phi) + V_{\text{glue}}(L, \bar{L}) \right\}, \quad (1)$$

with the shorthand notation $\int_x = \int_0^\beta dx_0 \int d^3x$ and $\beta = 1/T$, where T stands for the temperature. The subscript k in Γ_k indicates that an infrared (IR) cutoff is applied to the effective action, such that quantum and thermal fluctuations of momenta $p \lesssim k$ are suppressed. The full effective action is resolved as $k \rightarrow 0$, and thus k plays a role as the renormalization group (RG) scale. The field $\Phi = (q, \bar{q}, \phi)$ includes the three-flavor quark field $q = (q_u, q_d, q_s)^\top$ and the scalar and pseudoscalar meson fields $\phi = (\sigma, \pi)$. The mesons are in the adjoint representation of the $U(N_f = 3)$ group, which reads

$$\Sigma = T^a (\sigma^a + i\pi^a), \quad a = 0, 1, \dots, 8, \quad (2)$$

with

$$T^0 = \frac{1}{\sqrt{2N_f}} \mathbb{1}_{N_f \times N_f}, \quad (3)$$

and

$$T^a = \frac{\lambda^a}{2} \quad (a = 1, \dots, 8), \quad (4)$$

where λ^a are the Gell-Mann matrices. The quark and meson fields interact with each other through the Yukawa coupling h_k with

$$\Sigma_5 = T^a (\sigma^a + i\gamma_5 \pi^a). \quad (5)$$

The quark chemical potential μ in [\(1\)](#) is related to the baryon chemical potential μ_B with $\mu = \mu_B/3$, where other chemical potentials, e.g., the chemical potentials for the electric charge and strangeness, are assumed to be vanishing.

The mesonic potential in [\(1\)](#), i.e., the matter sector of the effective potential, reads

$$V_{\text{matt},k}(\phi) = \tilde{V}_k(\rho_1, \rho_2) - c_A \xi - c_l \sigma_l - c_s \sigma_s, \quad (6)$$

with

$$\rho_1 = \text{tr}(\Sigma \cdot \Sigma^\dagger), \quad (7)$$

$$\rho_2 = \text{tr} \left(\Sigma \cdot \Sigma^\dagger - \frac{1}{3} \rho_1 \mathbb{1}_{3 \times 3} \right)^2, \quad (8)$$

where ρ_1 and ρ_2 are invariant under the transformations $SU_V(3) \times SU_A(3) \times U_V(1) \times U_A(1)$ in the flavor space. Here σ_l and σ_s indicate the scalar mesons of light and strange quarks, respectively. The relevant strength constants c_l and c_s result in explicit breaking of the chiral symmetry, as well as the breaking of the flavor symmetry from the three-flavor case to that of 2+1 flavors. The $U_A(1)$ symmetry is broken by the Kobayashi-Maskawa-'t Hooft determinant, viz.,

$$\xi = \det(\Sigma) + \det(\Sigma^\dagger), \quad (9)$$

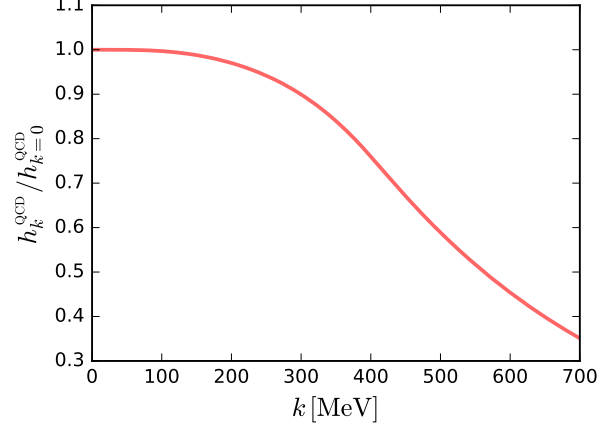


FIG. 1. Yukawa coupling, normalized to unity for $k = 0$, as a function of the RG scale k , computed from the first-principles QCD within the fRG in the vacuum [2].

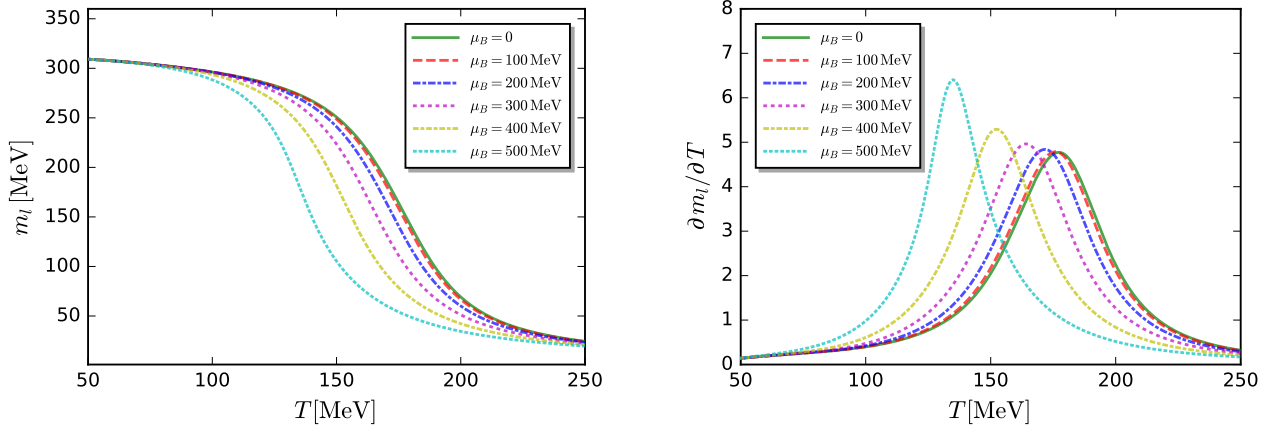


FIG. 2. Constituent light quark mass (left panel) and its derivative to the temperature (right panel) as functions of the temperature with several values of baryon chemical potential.

arising from quantum fluctuations, whose strength is controlled by the constant c_A .

The glue dynamics is encoded in the glue potential V_{glue} in (1), also known as the Polyakov loop potential. The Polyakov loop is related to the temporal gluon background field A_0 , that reads

$$L(\mathbf{x}) = \frac{1}{N_c} \langle \text{Tr } \mathcal{P}(\mathbf{x}) \rangle, \quad \bar{L}(\mathbf{x}) = \frac{1}{N_c} \langle \text{Tr } \mathcal{P}^\dagger(\mathbf{x}) \rangle, \quad (10)$$

with

$$\mathcal{P}(\mathbf{x}) = \mathcal{P} \exp \left(ig \int_0^\beta d\tau A_0(\mathbf{x}, \tau) \right), \quad (11)$$

where $\mathcal{P}$ on the right side denotes the path ordering, and g is the strong coupling constant.

In this work, we employ the Haar glue potential [1, 3],

$$V_{\text{glue}}(L, \bar{L}) = T^4 \bar{V}_{\text{glue-Haar}}, \quad (12)$$

with

$$\bar{V}_{\text{glue-Haar}} = -\frac{\bar{a}(T)}{2} \bar{L}L + \bar{b}(T) \ln M_H(L, \bar{L}) + \frac{\bar{c}(T)}{2} (L^3 + \bar{L}^3) + \bar{d}(T)(\bar{L}L)^2, \quad (13)$$

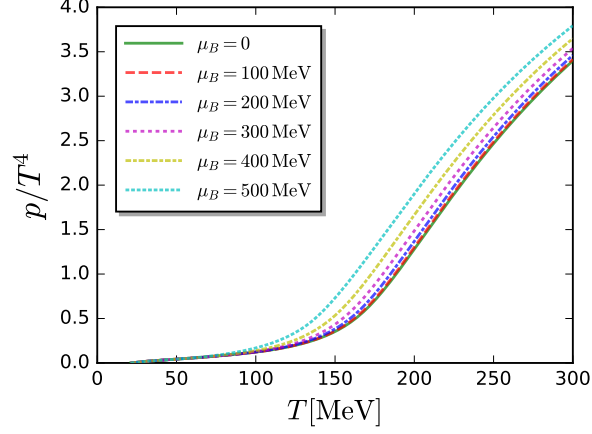


FIG. 3. Pressure normalized by T^4 as a function of the temperature at baryon chemical potential $\mu_B = 0, 100, 200, 300, 400$ and 500 MeV. These results are obtained in the 2+1 flavor LEFT.

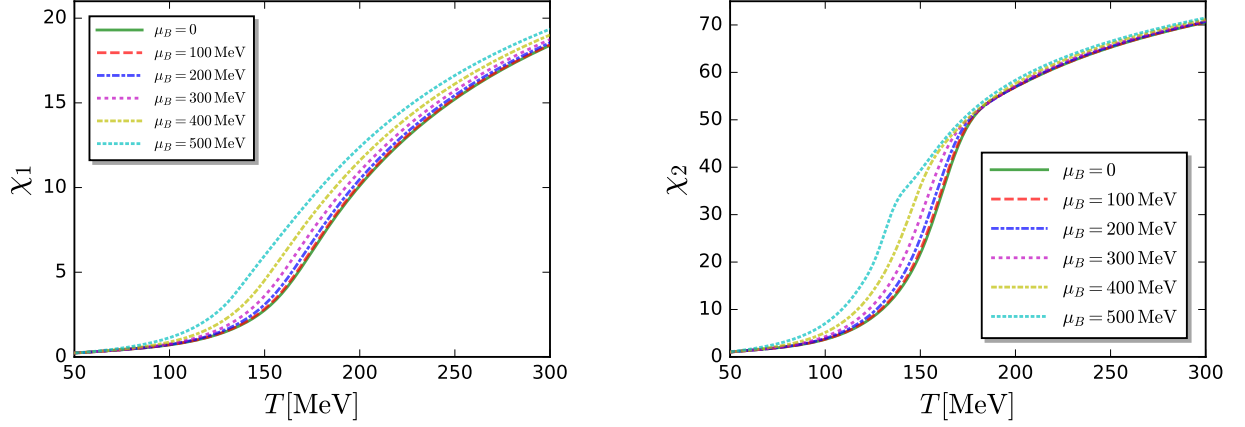


FIG. 4. Dimensionless entropy (left panel) and heat capacity (right panel) normalized by appropriate powers of T , i.e., χ_1 and χ_2 , as functions of the temperature with several values of baryon chemical potential.

where the Haar measure reads

$$M_H(L, \bar{L}) = 1 - 6\bar{L}L + 4(L^3 + \bar{L}^3) - 3(\bar{L}L)^2. \quad (14)$$

The temperature dependence of the coefficients $\bar{a}$, $\bar{c}$, $\bar{d}$ in (13) is parameterized as

$$x(T) = \frac{x_1 + x_2/(t+1) + x_3/(t+1)^2}{1 + x_4/(t+1) + x_5/(t+1)^2}, \quad x \in (\bar{a}, \bar{c}, \bar{d}), \quad (15)$$

and that of $\bar{b}$ as

$$\bar{b}(T) = \bar{b}_1(t+1)^{-\bar{b}_4} \left(1 - e^{\bar{b}_2/(t+1)^{\bar{b}_3}}\right). \quad (16)$$

Here t in (15) and (16) is the reduced temperature, that is

$$t = \alpha(T - T_c^{\text{glue}})/T_c^{\text{glue}}, \quad (17)$$

where the parameters $T_c^{\text{glue}} = 250$ MeV and $\alpha = 0.6$ are used throughout this work. The values of other parameters in (15) and (16) can be found in [1, 3].

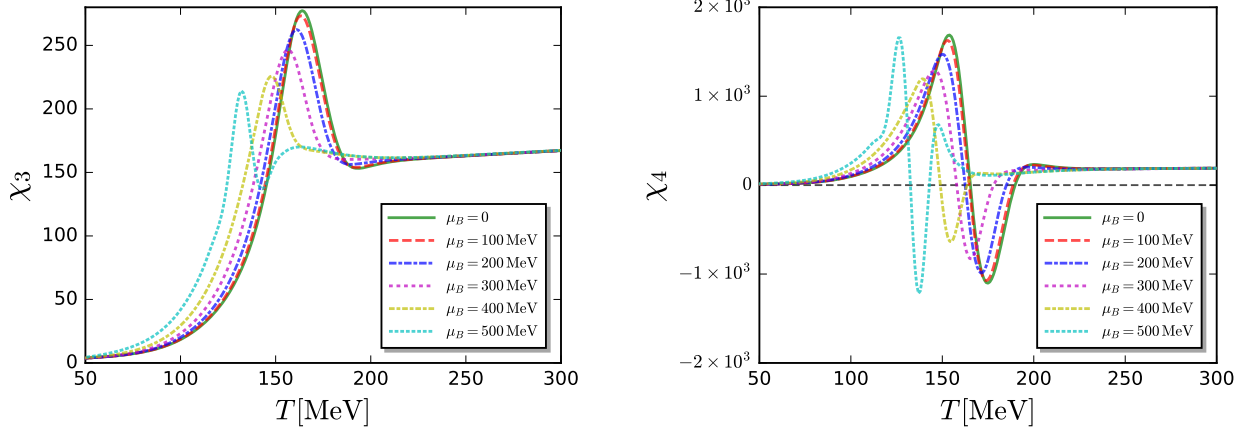


FIG. 5. Skewness (left panel) and kurtosis (right panel) of the entropy fluctuations, i.e., χ_3 and χ_4 , as functions of the temperature with several values of baryon chemical potential.

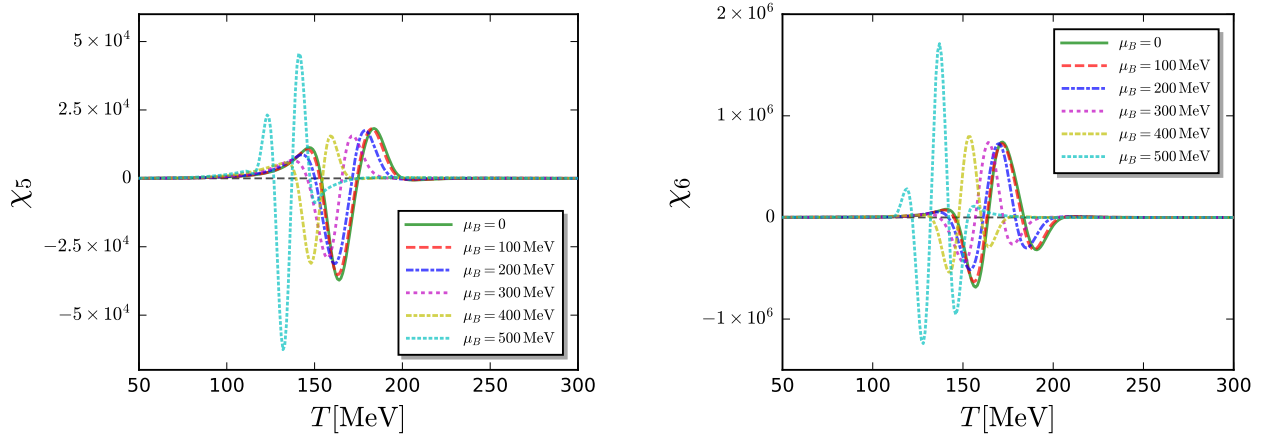


FIG. 6. Fifth (left panel) and sixth (right panel) order fluctuations of the entropy, i.e., χ_5 and χ_6 , as functions of the temperature with several values of baryon chemical potential.

Moreover, using the same method in [4], we employ the dependence of the Yukawa coupling on the RG scale k calculated from the first-principles QCD [2], as an input for the LEFT. Then the Yukawa coupling in the LEFT now reads

$$h_k = h_0 \frac{h_k^{\text{QCD}}}{h_{k=0}^{\text{QCD}}}, \quad (18)$$

where h_k^{QCD} is computed by using the fRG approach to the first-principles QCD in the vacuum [2], as shown in Fig. 1. Here the parameter in the LEFT $h_0 = 12$ is determined by fitting the constituent light u and d quark mass $m_l = 311$ MeV. Furthermore, we use the same values of parameters in the matter sector in (6) as those in [1].

In the left panel of Fig. 2, we show the constituent masses for the u , d light quarks calculated in the 2+1 flavor LEFT, depicted as functions of the temperature at several values of μ_B . Their respective derivatives with respect to the temperature are shown in the right panel of Fig. 2, from which one can determine the pseudo-critical temperature for the chiral crossover through the location of the peak. Figure 3 displays the temperature dependence of pressure calculated in our 2+1 flavor LEFT-fRG framework for baryon chemical potential μ_B ranging from $\mu_B = 0$ to 500 MeV, from which one can compute the temperature derivatives of pressure. The relevant results, from the first to sixth order derivatives, are presented in Figs. 4 to 6, which stand for the entropy and its fluctuations of different orders.

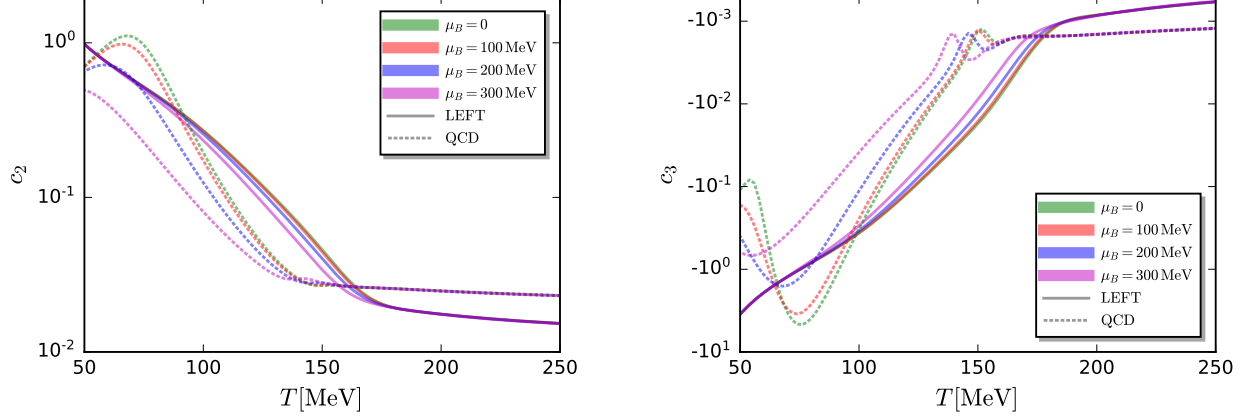


FIG. 7. Comparison between the LEFT and QCD computations for the second- and third-order fluctuations of temperature, i.e., c_2 (left panel) and c_3 (right panel), as functions of the temperature with several values of baryon chemical potential. The LEFT and QCD results are denoted by the solid and dashed lines, respectively.

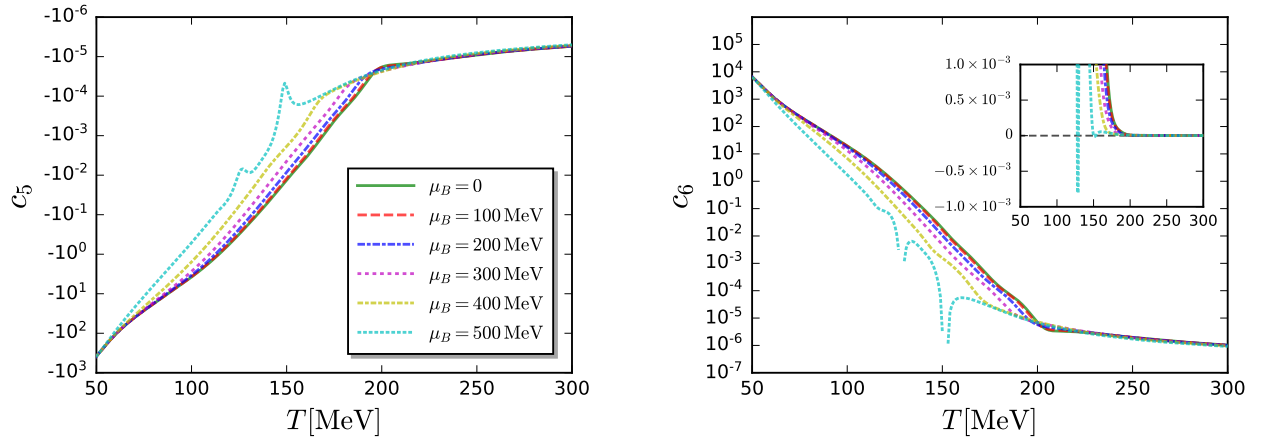


FIG. 8. Hyper-order fluctuations of temperature, c_5 (left panel) and c_6 (right panel), as functions of the temperature with several different values of baryon chemical potential. The inset on the right shows the plot by using the linear y -axis, where the zero-crossing for c_6 at high baryon chemical potential is clear.

S.2. Preliminary results of temperature fluctuations calculated in QCD within fRG

In addition, we also compare the temperature fluctuations obtained in the LEFT with the preliminary results obtained in QCD within the fRG, whose setup is detailed in [2]. The relevant results are presented in Fig. 7. One can see that although there are some differences, the variance and skewness of temperature fluctuations calculated from the LEFT and QCD are consistent with each other qualitatively. Note that the main conclusions obtained from the calculations in LEFT are corroborated by the QCD results, e.g., the skewness of temperature fluctuations is negative and the variance decreases significantly with the increase of temperature.

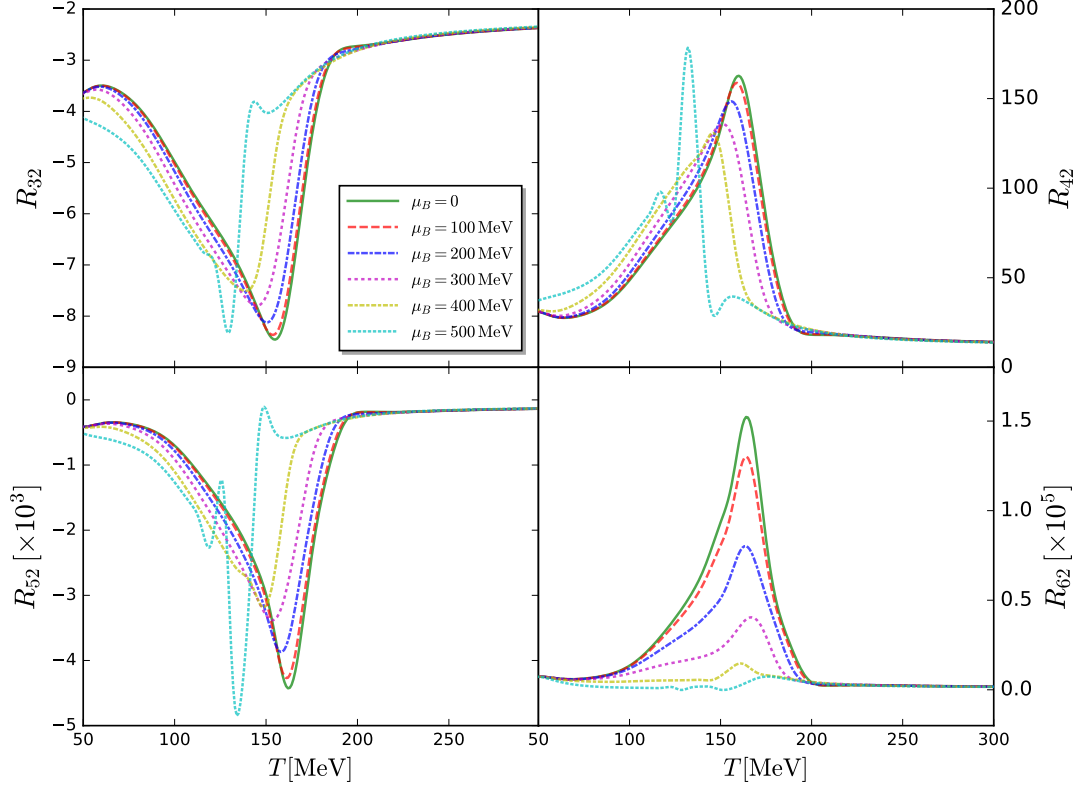


FIG. 9. Ratios between high-order temperature fluctuations and the variance, $R_{32} = c_3/c_2^2$, $R_{42} = c_4/c_2^3$, $R_{52} = c_5/c_2^4$, $R_{62} = c_6/c_2^5$, as functions of the temperature with several different values of baryon chemical potential.

S.3. Hyper-order fluctuations of temperature

The fifth- and sixth-order fluctuations of temperature are given by

$$\begin{aligned}
 c_5 = T^{11} & \left[-15 \left(\frac{\partial^2 p}{\partial T^2} \right)^{-7} \left(\frac{\partial^3 p}{\partial T^3} \right)^3 - \left(\frac{\partial^2 p}{\partial T^2} \right)^{-5} \frac{\partial^5 p}{\partial T^5} + 10 \left(\frac{\partial^2 p}{\partial T^2} \right)^{-6} \frac{\partial^3 p}{\partial T^3} \frac{\partial^4 p}{\partial T^4} \right] \\
 c_6 = T^{14} & \left[105 \left(\frac{\partial^2 p}{\partial T^2} \right)^{-9} \left(\frac{\partial^3 p}{\partial T^3} \right)^4 - 105 \left(\frac{\partial^2 p}{\partial T^2} \right)^{-8} \left(\frac{\partial^3 p}{\partial T^3} \right)^2 \frac{\partial^4 p}{\partial T^4} + 10 \left(\frac{\partial^2 p}{\partial T^2} \right)^{-7} \left(\frac{\partial^4 p}{\partial T^4} \right)^2 \right. \\
 & \left. + 15 \left(\frac{\partial^2 p}{\partial T^2} \right)^{-7} \frac{\partial^3 p}{\partial T^3} \frac{\partial^5 p}{\partial T^5} - \left(\frac{\partial^2 p}{\partial T^2} \right)^{-6} \frac{\partial^6 p}{\partial T^6} \right].
 \end{aligned} \tag{19}$$

The relevant numerical results are shown in Fig. 8.

S.4. Ratios of temperature fluctuation cumulants

In relativistic heavy-ion collisions, the event-averaged mean transverse momentum $\langle p_T \rangle$ of charged particles exhibits an approximate linear dependence on the system temperature, $\langle p_T \rangle = aT$. The parameter a represents the proportionality coefficient. In order to eliminate the influence from this coefficient that is not determined quite well, we instead analyze dimensionless ratios of temperature fluctuation cumulants:

$$R_{32} = \frac{c_3}{c_2^2}, \quad R_{42} = \frac{c_4}{c_2^3}, \quad R_{52} = \frac{c_5}{c_2^4}, \quad R_{62} = \frac{c_6}{c_2^5}. \tag{20}$$

where the powers of the variance in the denominators are chosen to balance the powers of T . The relevant ratios, presented in Fig. 9, reveal that the cumulant ratios develop an increasingly rich nonmonotonic structure while exhibiting systematically reduced amplitudes with the increase of μ_B , reflecting competing effects where enhanced critical fluctuations near the sharpened phase boundary emerge concurrently with the overall suppression of the magnitude of temperature fluctuation.

S.5. Relation between the temperature T and the mean transverse momentum $\langle p_T \rangle$

Considering a free relativistic gas in thermal equilibrium at temperature T with Boltzmann statistics, where the mass of particle, e.g., the pion $m_\pi \sim 150$ MeV, is negligible in comparison to the typical value of the transverse momentum $p_T \sim 1$ GeV, the particle number reads

$$N = (2s + 1)V \int \frac{d^3p}{(2\pi)^3} e^{-p/T}, \quad (21)$$

where s denotes the spin of the particle and V the volume of system. One readily finds for the mean transverse momentum of particles

$$\langle p_T \rangle = \frac{1}{N} (2s + 1)V \int \frac{d^3p}{(2\pi)^3} p_T e^{-p/T}, \quad (22)$$

where the momentum and the transverse momentum is related through

$$p = \sqrt{p_T^2 + p_z^2}, \quad (23)$$

with the longitudinal momentum p_z . Combining Eqs. (21) and (22) we have

$$\langle p_T \rangle = \frac{\int \frac{d^3p}{(2\pi)^3} p_T e^{-p/T}}{\int \frac{d^3p}{(2\pi)^3} e^{-p/T}}. \quad (24)$$

From the dimensional analysis in Eqs. (24), one immediately arrives at

$$\langle p_T \rangle = a T, \quad (25)$$

that is, $\langle p_T \rangle$ is linearly proportional to the temperature, with the proportionality coefficient a . Note that this is different from the non-relativistic case, where it is easy to find that $\langle p_T \rangle \sim T^{1/2}$. In order to determine the coefficient a in Eq. (25), one has to compute the expression in Eq. (24). The integral in the numerator in Eq. (24) cannot be computed analytically, but one can expand the momentum in the exponent in powers of p_z/p_T , i.e.,

$$\begin{aligned} p &= p_T \left(1 + \frac{p_z^2}{p_T^2} \right)^{1/2} \\ &= p_T + \frac{1}{2} \frac{p_z^2}{p_T} + \dots \end{aligned} \quad (26)$$

Expanding Eq. (26) to the second order, and substituting into Eq. (24), one finds

$$\langle p_T \rangle \approx \frac{15}{16\sqrt{2}} \pi T \approx 2.08 T. \quad (27)$$

Note that this value is close to the exact value

$$a = 2.356, \quad (28)$$

obtained by directly computing Eq. (24) numerically.

[1] R. Wen, C. Huang, and W.-J. Fu, Baryon number fluctuations in the 2+1 flavor low energy effective model, [Phys. Rev. D](#) **99**, 094019 (2019), [arXiv:1809.04233 \[hep-ph\]](#).

- [2] W.-j. Fu, J. M. Pawłowski, and F. Rennecke, QCD phase structure at finite temperature and density, [Phys. Rev. D **101**, 054032 \(2020\)](#), [arXiv:1909.02991 \[hep-ph\]](#).
- [3] P. M. Lo, B. Friman, O. Kaczmarek, K. Redlich, and C. Sasaki, Polyakov loop fluctuations in SU(3) lattice gauge theory and an effective gluon potential, [Phys. Rev. **D88**, 074502 \(2013\)](#), [arXiv:1307.5958 \[hep-lat\]](#).
- [4] W.-j. Fu, X. Luo, J. M. Pawłowski, F. Rennecke, and S. Yin, Ripples of the QCD critical point, [Phys. Rev. D **111**, L031502 \(2025\)](#), [arXiv:2308.15508 \[hep-ph\]](#).